ECE 308 Lab 5: Modeling Inverted Pendulum

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1. Introduction

This lab was focused on designing and implementing a control system for a rotary inverted pendulum. Specifically, we are interested in stabilizing the pendulum in the inverted position. This was done by first modeling the system and determining the linearized differential equations of motion. Then, after setting up the inverted pendulum, we focused on balance control design. To design a state feedback controller to balance the pendulum upright, we calculated the poles necessary to reach the given damping ratio and undamped natural frequency criteria, and then used those to determine the gain matrix K. Then, after implementing the balance control, we worked on the swing-up control. This was done by first obtaining real-time energy of the pendulum, then building a controller subsystem based on that. Then, we converted acceleration to servo voltage, and then manually tuned the proportional gain until the swing-up control was functional.

2.Results

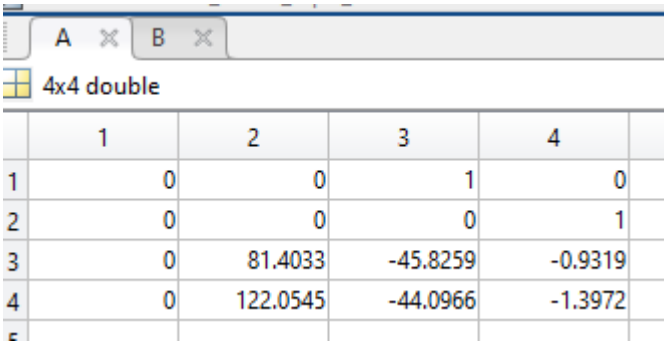
Part a: System Modeling

Question 5.9

$$\begin{matrix}\theta' \\ \alpha' \\ \theta'' \\ \alpha''\end{matrix} = \begin{matrix}0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & \frac{1}{4}m_p^2L_p^2L_rg * \frac{1}{J_T} & \frac{1}{J_T}\left(-\left(J_p+\frac{1}{4}m_pL_p^2\right)D_r\right) & \frac{1}{J_T}\left(-\frac{1}{2}m_pL_pL_rD_p\right) \\ 0 & \frac{\frac{1}{2}m_pL_pg(J_r+m_pL_r^2)1}{J_T} & \frac{1}{J_T}\left(-\frac{1}{2}m_pL_pL_rD_r\right) & \frac{1}{J_T}\left(-J_r+m_pL_r^2\right)D_p\end{matrix} * \begin{matrix}\theta \\ \alpha \\ \theta' \\ \alpha'\end{matrix} + * \begin{matrix}\theta \\ \alpha \\ \theta' \\ \alpha'\end{matrix}$$

$$A = \begin{matrix}0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & \frac{1}{4}m_p^2L_p^2L_rg * \frac{1}{J_T} & \frac{1}{J_T}\left(-\left(J_p+\frac{1}{4}m_pL_p^2\right)D_r\right) & \frac{1}{J_T}\left(-\frac{1}{2}m_pL_pL_rD_p\right) \\ 0 & \frac{\frac{1}{2}m_pL_pg(J_r+m_pL_r^2)1}{J_T} & \frac{1}{J_T}\left(-\frac{1}{2}m_pL_pL_rD_r\right) & \frac{1}{J_T}\left(-J_r+m_pL_r^2\right)D_p\end{matrix}$$

$$B = \begin{matrix}0 \\ 0 \\ \left(J_p+\frac{1}{4}m_pL_p^2\right) * \frac{1}{J_T} \\ \left(\frac{1}{2}m_pL_pL_r\right) * \frac{1}{J_T}\end{matrix}$$

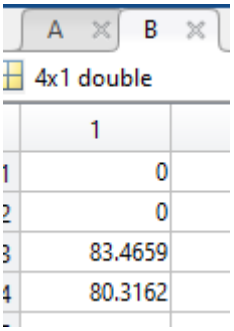


A | B

4x4 double

	1	2	3	4
1	0	0	1	0
2	0	0	0	1
3	0	81.4033	-45.8259	-0.9319
4	0	122.0545	-44.0966	-1.3972

Matrix A



A | B

4x1 double

	1
1	0
2	0
3	83.4659
4	80.3162

Matrix B

ii.

Open loop poles = eigenvalues of A or det (Is – A) = 0. Using eig(A) in MATLAB returned:

```
ans =

         0
  -48.4156
    7.0569
   -5.8644
```

Part B: Balance Control

i.

Question 5.10

ζ = 0.7, natural damping freq = 4, s = -30, -40

s^2 + 2ζwns + wn^2 = 0

s^2 + 5.6s + 16 = 0

à s = -2.8 $\pm$ 2.86i

s = -2.8 + 2.86i, -2.8 – 2.86i, -30, -40

ii.

Question 5.11

```
Balance control gain:
   -5.2612   28.1569   -2.7576    3.2190
```

K values (5.11):

iii.

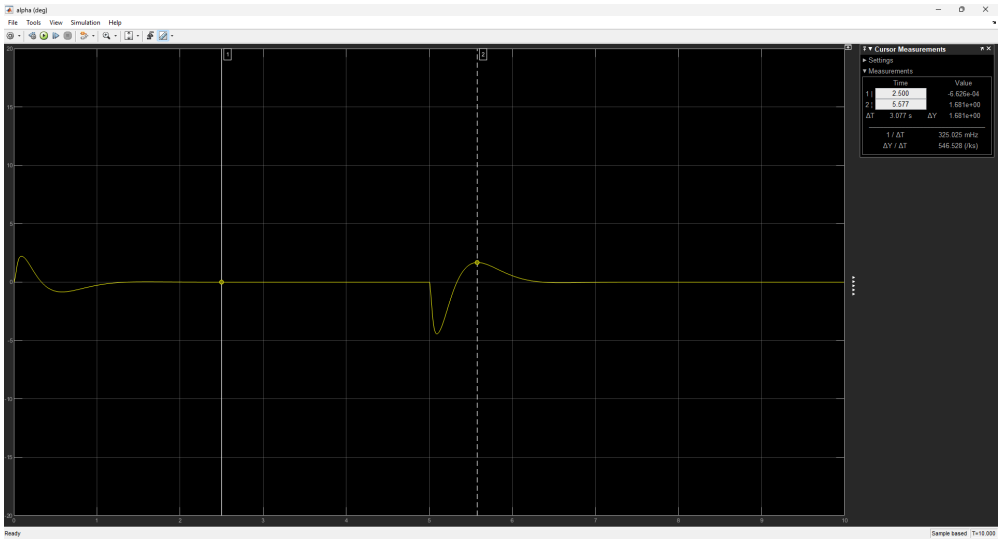


Figure 1: Alpha(deg) Simulation

Figure 1 shows a scope capture of the simulated of the angle of the balance control, with 0 being the positive z-axis. From this waveform we can assume the arm will oscillate with reducing magnitudes if it is moved from the positive z-axis.

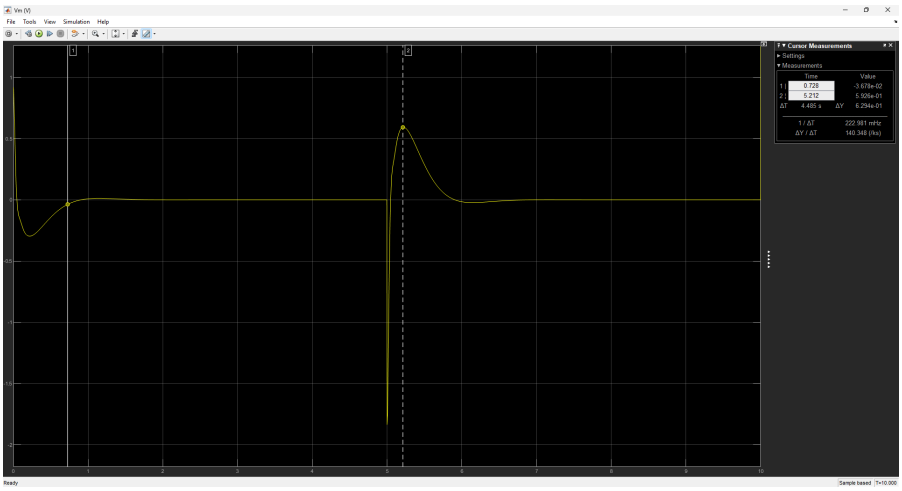


Figure 2: Vm (Volts) Simulation

This figure display the Voltage needed to stay stable, this waveform algins with the alpha simulation as an initial high negative voltage and then a gradually decreasing oscillation is very similar to the movement waveform.

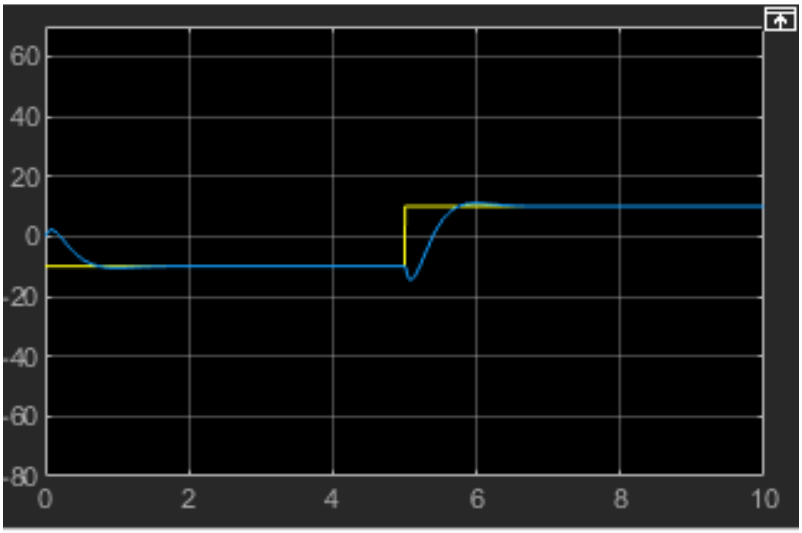


Figure 3: Theta (deg) Simulation

The blue signal on the scope capture for figure 3 demonstrates the expected horizontal rotation when the system is turned on with a step input. The typical oscillation with decreasing frequency is clearly visible and aligns with all other waveforms.

	Simulation	Hardware
|Vm| max	5.2V	1.3
$\|\alpha\|$max	5.577deg	3.2

Table 5.1: Balance Control

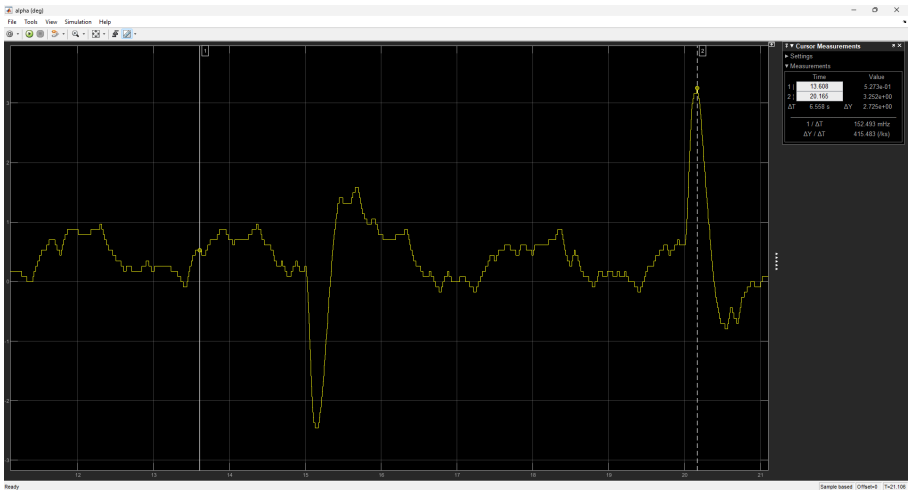


Figure 4: Alpha(deg) Hardware Implementation

While much less clean than the simulation, this wave also shows an oscillating waveform with decreasing magnitude, so this scope capture supports the idea that this controller is working as intended.



Figure 5: Theta(deg) Hardware Implementation

Unfortunately, this figure was not fully captured by the scope and as such determining whether the horizontal rotation was as expected cannot be done with it. However, due to it physically working there are no reasons we would assume it was wrong.

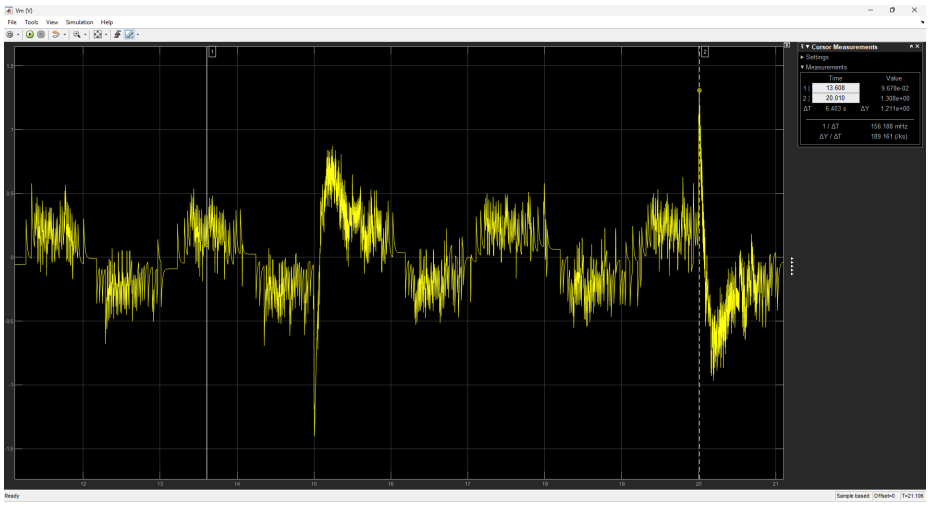


Figure 6: Hardware implementation Vm(V)

In the final hardware implementation scope, the reducing in magnitude oscillations are again clearly visible, and align with the alpha waveform.

Part C: Swing Up Control

This section of the lab was worked on as a group with x and x, as the member of our group with the code from the previous section was out of class for an excused absence.

Er = mgLp(1-cosa) = 0.42

Controller gain: μ = 3.3

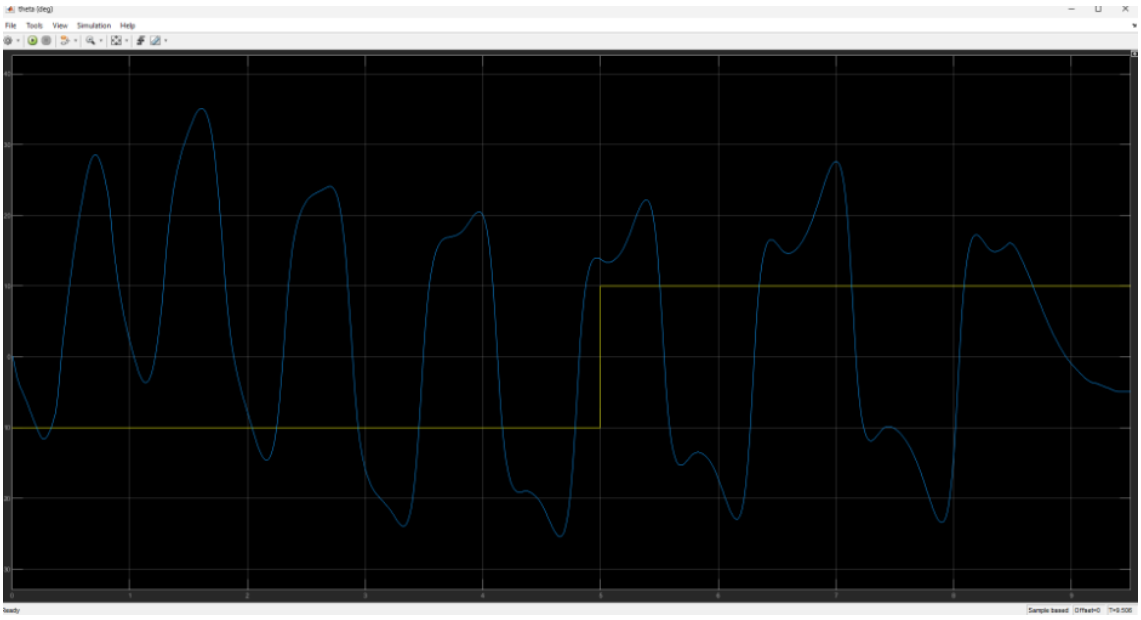


Figure 7: Swing-up rotary arm angle (Θ)

This scope capture shows the blue waveform of the horizontal angle movement of the servo motor. A gradual decline in magnitude is noticeable which makes sense as the motor initially moves a lot back and forth and then reaches an equilibrium point and moves much less.

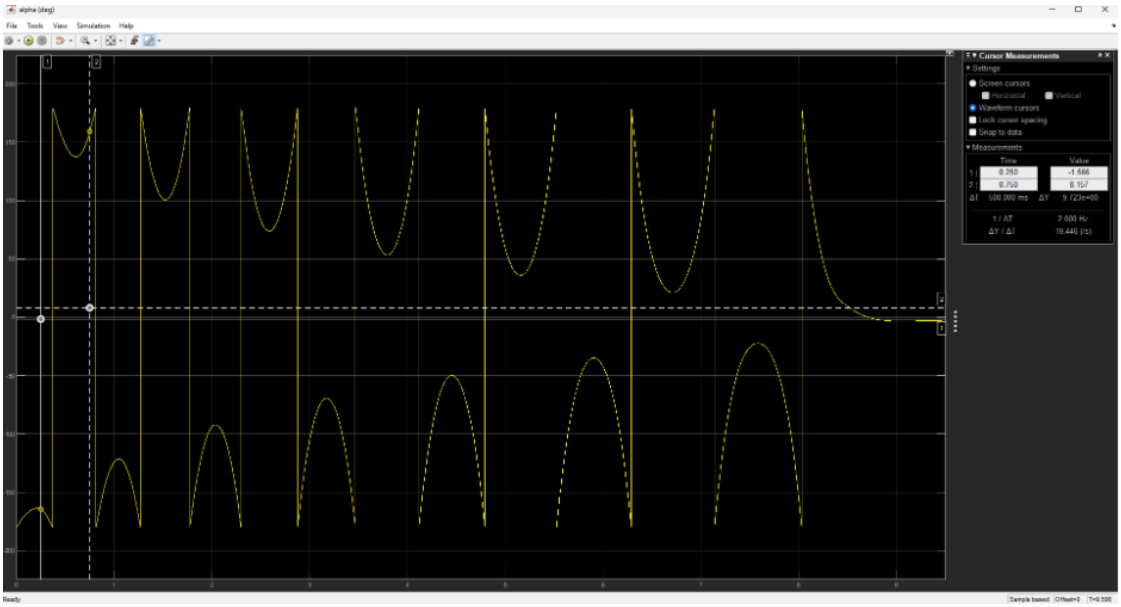


Figure 8: Swing-up pendulum angle (α)

This scope capture shows the angle of the pendulum rod, which is the other axis of rotation on the system. The big jumps from -180 to +180 are the rod passed the point directly below the z axis, but when taking that into account the waveform becomes a sin wave with increasing magnitude. However, that magnitude does max out when it reaches the z axis and stays a lot more constant, holding up the pendulum.

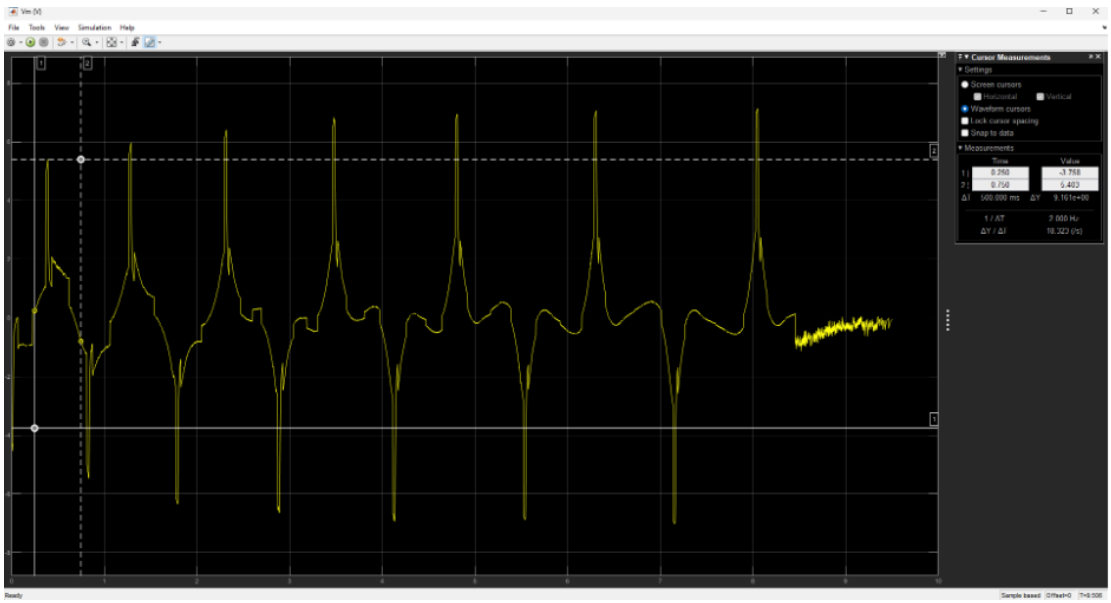


Figure 9: Swing-up control input (V_m)

In this scope capture, the yellow waveform in V shows the input voltage to keep the pendulum up, at first, it jolts back and forth which shows it supplying a negative and positive voltage to move the pendulum back and forth. This is all relatively consistent until it becomes stationary at its pivot point and provides a much smaller stabilizing voltage.

2. Conclusions

Our experimental results showed that the controller using the state-space matrices effectively balanced the inverted pendulum. This is shown in our results from Table 5.1, with both the simulation and hardware producing mostly expected results. From part aii, initially there was an open loop pole at s = 7.0569, which would make the system unstable, but as shown in bi, the gain matrix K stabilized the system, as none of the closed loop poles are in the RHP.

The Swing up Controller worked perfectly, when turned on it would start rocking back and forth as the scope captures show. Gradually its swing would get larger until it reached the peak and stopped right there. When putting some force on the pendulum, it would provide its own force back to stay upright. The main challenges faced during the programming of the controller arised from the large size of the Simulink model and making small errors that would make the pendulum swing uncontrollably.